Kernel-Checked Opening Theory and Coefficient-Sensitive Newton Geometry of a Six-State Dark Quantum Channel

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Independent research preprint

Version R2.1 — 10 August 2026
Repository and reproducibility materials

Abstract

Exact zero transfer in a finite quantum channel is a statement about every time, whereas the loss of one symmetry certificate concerns only one proof. This paper gives a kernel-checked local opening theory for a locked six-state real Hamiltonian under internal diagonal perturbations. In Lean 4 with Mathlib, we formalise the signed involutions and all-depth baseline darkness, the unreduced selected cofactor, the pointwise dark-plane decomposition, endpoint word support, the complete fixed-ray trichotomy, the fixed-fibre Newton theorem and the analytic bridge to the actual matrix exponential. The selected resolvent numerator is $L(z^2 + 2z) + 2Q(z + 1) + R$, where $L = d_1 - d_2 + d_3 - d_4$, $Q = d_2d_4 - d_1d_3$, and $R = d_1d_2(d_3 - d_4) + d_3d_4(d_1 - d_2)$. The formal development proves that the two fixed non-dark fibres have Newton polyhedra $(1, 3) + \mathbb{R}^2_{\geq 0}$ and $(2, 4) + \mathbb{R}^2_{\geq 0}$, while the dark fibre is empty, and identifies these coefficients with the convergent amplitude series. Beyond that verified spine, conventional exact proofs establish monomial-fibre aggregation, a moving-arc lower-hull classification, a four-vertex universal Newton polyhedron, a 16-cone Gröbner fan and a fixed-weight Noetherian termination theorem. The paper marks this verification boundary explicitly: no effective uniform depth bound, finite global recursive atlas, or blanket formalisation claim is made.

Evidence convention. Claims carrying an explicit status use only PROVED, KERNEL-CHECKED, VERIFIED-EXACT, VERIFIED-BYTE, MEASURED with a sample size, FALSIFIED, CONJECTURE, or UNAVAILABLE. KERNEL-CHECKED means that the stated Lean theorem and its dependencies were accepted by the pinned Lean kernel; it does not extend to nearby results not named in the traceability map. A computed identity is never promoted to a theorem without a proof, and a failed certificate is never called an opening.

Contents

1 Introduction

Dark states and destructive interference are standard mechanisms in coherent population trapping and adiabatic transfer (Aspect et al., 1988; Gaubatz et al., 1990). Recent work gives broad structural descriptions of dark states in multilevel systems (Zhao, Kuang and Liao, 2026); continuous-time quantum-walk research likewise studies exact or near suppression of transfer (Godsil, 2012; Sett et al., 2019), while time-reversal breaking is known to alter quantum transport (Zimborás et al., 2013). Those settings motivate, but do not answer, the local algebraic question studied here: once a channel is exactly dark, which coefficient is the first to survive under a declared diagonal perturbation, and how does that answer change under multi-scale substitution?

The central contribution is not only an exact answer for one locked channel, but a machine-checked proof of its fixed-ray and fixed-fibre spine. The model, signed-symmetry argument, determinant identity, word recurrence, coefficient support, branch classification, convex-geometric conclusion and matrix-exponential bridge have been formalised in Lean 4 (de Moura and Ullrich, 2021) using Mathlib (The mathlib Community, 2020). This changes the evidentiary architecture of the paper: exact scripts are retained as independent checks, while the main finite-algebra chain is anchored to named kernel-checked theorems. The precise map and trust boundary appear in section 2.

The companion Paper I proves that, for a finite Hamiltonian H, exact zero transfer from e_0 to e_5 is equivalent to the vanishing of all moments $e_5^\top H^k e_0$, to orthogonality of e_5 to the source Krylov space, and to the corresponding spectral-projector conditions (Medford, 2026b). Signed involutions provide sufficient exclusion certificates there. This paper starts from those equivalences and develops the coefficient-sensitive geometry of one six-state channel. It does not repeat the graph atlas, magnetic-flux examples, norm bounds, mass estimates or open-system extensions of Paper I.

The first fixed-direction version appeared in the author-supplied R1.2 manuscript (Medford, 2026a). The present version retains its exact numerator and trichotomy, supplies the complete Lean formalisation through the analytic bridge, repairs the scope of moving-direction cancellation, and adds endpoint words, pullback aggregation, analytic arcs, universal coordinates, Gröbner/Rees data and the fixed-weight Noetherian termination theorem.

1.1 Contribution and evidence structure

Resolvents and Markov parameters are standard in realisation and transfer theory (Ho and Kalman, 1966; Noferini et al., 2023); noncommutative word series are standard in nonlinear systems and weighted automata (Berstel and Reutenauer, 2010; Fliess, 1981; Sakarovitch, 2009). Newton polyhedra, tropical initial forms, Gröbner fans, Rees algebras and arc contact orders also have mature theories (Ein, Lazarsfeld and Mustaţă, 2004; Fukuda, Jensen and Thomas, 2007; Kouchnirenko, 1976; Maclagan and Sturmfels, 2015; Varchenko, 1976). The contribution is their exact synthesis for a channel in which moment and amplitude faces can genuinely disagree.

The KERNEL-CHECKED contribution comprises:

(i) two explicit signed involutions and all-depth baseline darkness;

(ii) the exact unreduced three-coefficient numerator and its coefficient syzygy;

(iii) the pointwise union of two dark planes, each with a persistent signed certificate;

(iv) the locked endpoint-access support theorem and signed boundary power sum;

(v) the complete linear/quadratic/exact-zero fixed-ray trichotomy;

(vi) the two fixed-fibre Newton orthants, empty dark fibre and unique positive-weight vertices; and

(vii) the globally convergent identification of the formal coefficient array with the selected matrix-exponential amplitude.

The subsequent, conventionally proved extension comprises monomial-fibre aggregation, the analytic-arc lower hull and field-corrected tie rule, a universal four-vertex Newton polyhedron with 15 positive normal cells, a 16-cone Gröbner fan and the hierarchy from the dark ideal to a marked amplitude section, and a non-effective finite decision theorem for every fixed positive rational weight. These results are supported by exact symbolic and combinatorial certificates, but are not labelled KERNEL-CHECKED.

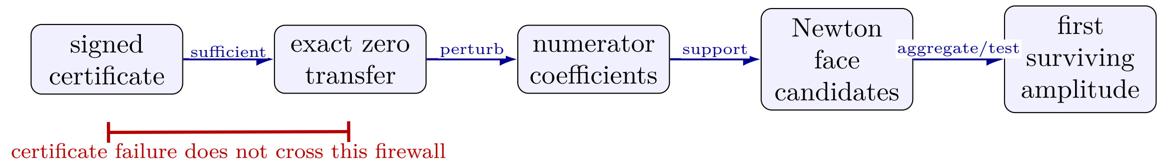

Figure 1: The logical chain. The signed-certificate-to-fixed-fibre route and its amplitude bridge are kernel-checked; later pullback and atlas constructions extend this core by conventional exact proofs. Every arrow is one-way unless an adjacent theorem states an equivalence.

1.2 Evidence and field conventions

All algebraic varieties are over a declared field. The locked physical family has real diagonal coordinates, while coefficient-torus cancellation is also studied over $\mathbb{C}$. Real non-cancellation conclusions are therefore never exported to complex leading constants. Named Lean theorems supply KERNEL-CHECKED claims; exact computation supplies VERIFIED-EXACT identities; conventional proofs supply PROVED claims. The source census mentioned later is MEASURED with its sample size. No numerical sampling establishes a universal statement.

Scope firewall. Closure is not a symmetry certificate; a broken certificate is not an opening; Newton support is not a surviving coefficient; and the first exponent p of the perturbation parameter is not the first time power q. Equally, a Lean proof of the fixed-fibre theorem is not a Lean proof of the later moving-arc or global-atlas theorems. These distinctions are maintained throughout.

2 Formal verification architecture and trust boundary

The fixed-ray and fixed-fibre core of this paper has been formalised in Lean 4, an interactive theorem prover with a small proof-checking kernel (de Moura and Ullrich, 2021), using the Mathlib library of formalised mathematics (The mathlib Community, 2020). The formalisation is not a translation of displayed formulas into numerical tests. It defines the locked six-state matrix, selected ports, perturbation, cofactor, word coefficients, coefficient support and real convex hull, and then constructs proof terms for the corresponding universal statements.

2.1 End-to-end verified spine

The organising dependency chain is recorded in table 1. Module names and theorem identifiers are part of the release interface: they identify the exact machine-checked statement rather than a nearby prose summary.

Stage	Paper-level claim	Lean module and principal theorem
M0–M1	Locked model, signed involutions and all-depth baseline darkness	`M3A.BaselineDarkness`; `baseline_moments_dark`
M2	Exact unreduced selected cofactor numerator	`M3A.SelectedNumerator`; `selectedNumerator_formula`
M3	Pointwise decomposition into the two dark planes and persistent certificates	`M3A.DarkPlanes`; `ell_quad_zero_iff_planes`
M4	Locked endpoint-access support and signed boundary power sum	`M3A.EndpointWords`; `sixState_support_bound`, `sixState_boundary_ray`
M5	Exact first amplitude coefficients and complete first-order protection on $L = 0$	`M3A.RayCoefficients`; `amplitudeCoeff_one_eq_zero_of_linear_zero`
M6	Exhaustive linear/quadratic/dark fixed-ray classification	`M3A.FixedRayTrichotomy`; `fixedRay_trichotomy`, `classifyRay_spec`
M7	Actual-support Newton orthants and unique positive-weight vertices	`M3A.FixedFibreNewton`; `fixedFibre_newton`
M8	Identification with the selected matrix-exponential amplitude	`M3A.AnalyticBridge`; `selectedRayAmplitude_eq_tsum_coefficients`

Table 1: The theorem-to-Lean dependency map. Each row depends on the rows above it; M8 closes the gap between finite word algebra and the analytic amplitude used in the paper.

The resulting chain is genuinely end to end. At each time depth q, M8 proves the finite identity

$$M_q(\varepsilon v) = \sum_{p=0}^{q} \varepsilon^p M_{p,q}(v),$$

where $M_{p,q}$ is the word coefficient defined in M4. It then proves global summability in t and the equality

$$K(\varepsilon v, t) = \sum_{q=0}^{\infty} \sum_{p=0}^{q} A_{p,q}(v)\varepsilon^p t^q, \qquad A_{p,q}(v) = \frac{(-i)^q}{q!} M_{p,q}(v). \tag{1}$$

Thus the support classified in M6 and convexified in M7 is the coefficient support of the actual matrix exponential, not an unconnected formal surrogate.

2.2 What the status does and does not mean

The release pins Lean and Mathlib at version 4.32.1. A clean build checks the root import through `M3A.AnalyticBridge`. Axiom inspection of every principal theorem reports only `propext`, `Classical.choice` and `Quot.sound`, the standard foundational axioms used by Lean and Mathlib. The project contains no `sorry`, `admit`, user-declared axiom or unsafe theorem dependency. Exact Python and computer-algebra scripts remain valuable independent regression checks, but they are not premises of the Lean proofs.

The formal boundary is equally important. M0–M8 cover the locked scalar-ray specialisation through the fixed-fibre Newton theorem and its analytic bridge. They do not yet formalise the fully general rectangular endpoint theorem, the ideal equality in a multivariate polynomial ring, monomial pullback, moving analytic arcs, the universal four-coordinate atlas, the Gröbner/Rees structures or the fixed-weight Noetherian theorem. Those later results retain their conventional proofs and exact-computation certificates. The transition is marked explicitly after section 8, and the detailed claim map is given in section F.

3 Locked model and observables

Let $e_0, \ldots, e_5$ be the standard basis of $\mathbb{C}^6$, with source e_0 and target e_5. The unperturbed real-symmetric Hamiltonian is

$$H_0 = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & -1 \\ 1 & 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & -1 \\ 0 & 1 & -1 & 1 & -1 & 0 \end{pmatrix}. \tag{2}$$

For $d = (d_1, d_2, d_3, d_4)$, set

$$D(d) = \operatorname{diag}(0, d_1, d_2, d_3, d_4, 0), \qquad H(d) = H_0 + D(d). \tag{3}$$

The fixed-ray specialisation is $d = \varepsilon v$, where $v \in \mathbb{R}^4$ is selected independently of ε. A moving direction $v = v(\eta)$ is a different, two-parameter object; it is not smuggled into a fixed-ray theorem.

Three observables must be distinguished:

$$G(z; d) = e_5^\top (zI - H(d))^{-1} e_0, \tag{4}$$

$$M_k(d) = e_5^\top H(d)^k e_0, \tag{5}$$

$$K(d, t) = e_5^\top e^{-itH(d)} e_0 = \sum_{k \geq 0} \frac{(-i)^k}{k!} M_k(d) t^k. \tag{6}$$

The numerator of G, the moment series and the amplitude are linked but not identical. In particular, the factor $(-i)^k/k!$ can change a face cancellation equation when different time depths share a pulled-back weight. The passage from the finite moment coefficients to this actual matrix exponential is itself part of the formal development: M8 proves the convergent identity equation (1) for every fixed complex ray, perturbation parameter and time.

Write

$$L = d_1 - d_2 + d_3 - d_4, \qquad Q = d_2 d_4 - d_1 d_3, \tag{7}$$

and

$$R = d_1 d_2 (d_3 - d_4) + d_3 d_4 (d_1 - d_2). \tag{8}$$

On a ray $d = \varepsilon(a, b, c, d)$ we write $\ell = a - b + c - d$ and retain $Q = bd - ac$; the letter R then denotes the cubic coefficient in equation (8), not a remainder, a Rees algebra, a ray parameter, or the cubic moment. This notation lock prevents the principal symbol collision in the source record.

Proposition 3.1 (Baseline darkness). *For $d = 0$, $K(0, t) = 0$ for every $t \in \mathbb{R}$, equivalently $M_k(0) = 0$ for every $k \geq 0$.*

Proof. Let S be either of the two signed involutions defined by the model. It satisfies $S^2 = I$, $SH_0 = H_0 S$, $Se_0 = e_0$ and $e_5^\top S = -e_5^\top$. Hence, for every k,

$$M_k(0) = e_5^\top H_0^k e_0 = e_5^\top S H_0^k S e_0 = -M_k(0),$$

so $M_k(0) = 0$ over characteristic zero. Termwise substitution in the globally convergent exponential series gives $K(0, t) = 0$. $\square$

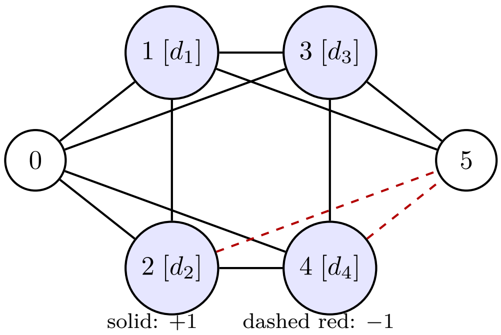

Figure 2: The locked six-state channel. Solid lines carry (+1), dashed red lines carry (-1), and the four internal vertices carry diagonal coordinates.

Formal status. The matrix identities, abstract signed-certificate implication and all-depth conclusion are KERNEL-CHECKED as `M3A.rOrig_involutive`, `M3A.rOrig_commutes_H0`, `M3A.selectedMoment_eq_zero_of_signedCertificate` and `M3A.baseline_moments_dark`. M8 checks the matrix-exponential passage; no finite moment cutoff is used as a proof.

Remark 3.2. *The proposition is about the channel, not about the persistence of any particular certificate. A perturbation may destroy a signed involution while the transfer remains exactly zero for another reason.*

4 Exact numerator and coefficient ideal

Let $\Delta(z; d) = \det(zI - H(d))$. The selected adjugate cofactor is denoted by $N(z; d)$, so $G = N/\Delta$ as a rational function before any common-factor reduction.

Theorem 4.1 (Exact selected numerator)**.** *For the locked model,*

$$\boxed{N(z; d) = L(z^2 + 2z) + 2Q(z + 1) + R.} \tag{9}$$

This identity is KERNEL-CHECKED *and independently* VERIFIED-EXACT*.*

Proof. Delete row 0 and column 5 from $zI - H(d)$, take the determinant and multiply by the cofactor sign $(-1)^{0+5}$. In the Leibniz expansion, collect terms by total diagonal degree. The degree-one terms are $Lz^2 + 2Lz$; the degree-two terms are $2Qz + 2Q$; the degree-three terms are R; higher degrees cancel. This gives equation (9). A second route expands $H(d)^k$, forms the Laurent series $G(z; d) = \sum_{k \geq 0} M_k(d) z^{-k-1}$, and multiplies by the independently computed denominator. The release script performs both integer routes and compares every monomial. □

Formal status. The direct cofactor construction, retained row and column embeddings, cofactor sign and universal identity over every commutative ring are checked by `M3A.selectedMinor_eq_explicit` and `M3A.selectedNumerator_formula`. The determinant proof uses no sampled evaluation or external symbolic output as a premise.

Proposition 4.2 (Two-generator coefficient ideal)**.** *Let $\mathbb{K}$ be a field of characteristic different from 2, equivalently a field in which 2 is invertible. In $A = \mathbb{K}[d_1, d_2, d_3, d_4]$, the ideal of the coefficients of N is*

$$I_N = (L, Q, R) = (L, Q). \tag{10}$$

Proof. The coefficient list of equation (9) contains L, $2Q$ and R, so invertibility of 2 gives $I_N = (L, Q, R)$. The integral syzygy

$$R = -(d_2 + d_4)Q - d_2 d_4 L. \tag{11}$$

shows that $R \in (L, Q)$, while $L, Q \in I_N$ because L and $2Q$ are coefficients and 2 is a unit. Hence equality. □

Formal status. The coefficient syzygy and the implication $L = Q = 0 \Rightarrow R = 0$ are KERNEL-CHECKED as `M3A.cubicNumeratorCoeff_syzygy` and `M3A.cubicNumeratorCoeff_eq_zero_of_linear_quadratic_zero`. The displayed equality of ideals, as an equality in a multivariate polynomial ring, remains a conventional algebraic proof rather than a theorem in the present Lean package.

The exact-zero locus of the selected rational entry is therefore the algebraic set $V(L, Q)$, subject to no inference from a reduced numerator alone: the identity is proved at the unreduced cofactor level. Potential numerator– denominator cancellation can lower a pole order, but cannot create a non-zero rational entry when the unreduced numerator is the zero polynomial.

Corollary 4.3 (Actual-coordinate transformed numerator)**.** *With $x = z^{-1}$,*

$$x^{-1}G(x^{-1}; d) = \frac{Lx^3(1 + 2x) + 2Qx^4(1 + x) + Rx^5}{\Delta^\sharp(x; d)}, \tag{12}$$

where $\Delta^\sharp(0; 0) = 1$. Hence the denominator is a unit in the formal local ring at $(x, d) = (0, 0)$.

5 Syzygy, dark planes and certificates

Define

$$P_A = \{d_1 = d_2,\ d_3 = d_4\}, \qquad P_B = \{d_1 = d_4,\ d_3 = d_2\}.$$

Theorem 5.1 (Pointwise dark-plane decomposition)**.** *Over every integral domain,*

$$L(d) = Q(d) = 0 \iff d \in P_A \cup P_B. \tag{13}$$

On P_A the original signed involution persists, and on P_B the alternative signed involution persists. Consequently every selected moment, and hence the selected amplitude, vanishes on the union.

Proof. Eliminate d_1 using $L = 0$, so $d_1 = d_2 - d_3 + d_4$. Then

$$Q = d_2 d_4 - (d_2 - d_3 + d_4)d_3 = (d_2 - d_3)(d_4 - d_3).$$

The absence of zero divisors gives either $d_2 = d_3$, $d_1 = d_4$, which is P_B, or $d_4 = d_3$, $d_1 = d_2$, which is P_A. The converse follows by substitution. On each plane the matching diagonal perturbation commutes with the stated signed involution, so the argument of proposition 3.1 applies at every perturbation value. □

Formal status. The equivalence equation (13) is KERNEL-CHECKED as `M3A.ell_quad_zero_iff_planes`; Lean proves it over every integral domain. The persistent commutators and the all-depth consequences are checked by `M3A.rOrig_commutes_hamiltonian_of_planeA`, `M3A.rAlt_commutes_hamiltonian_of_planeB` and `M3A.moments_dark_of_linear_quadratic_zero`. M8 then gives actual amplitude darkness.

Proposition 5.2 (Reduced ideal decomposition)**.** *Over any field of characteristic different from* 2*,*

$$(L, Q) = (d_1 - d_2, d_3 - d_4) \cap (d_1 - d_4, d_3 - d_2). \tag{14}$$

Consequently the exact-dark scheme is the reduced union of two two-dimensional linear planes.

Proof. The factorisation in the proof of theorem 5.1 gives the two distinct prime components after eliminating L. Pulling them back to the original polynomial ring gives equation (14); their intersection is radical. □

This scheme-theoretic ideal equality is PROVED in the paper but is outside the current Lean scope. Separating it from theorem 5.1 prevents the kernel-checked pointwise theorem from being silently upgraded to an unformalised statement about polynomial ideals.

On a fixed ray $v = (a, b, c, d)$, the two planes read

$$P_A: \ a = b, \ c = d, \qquad P_B: \ a = d, \ c = b. \tag{15}$$

They intersect in the scalar line $a = b = c = d$, an equality also checked in Lean by `M3A.onPlaneA_and_onPlaneB_iff_scalarLine`.

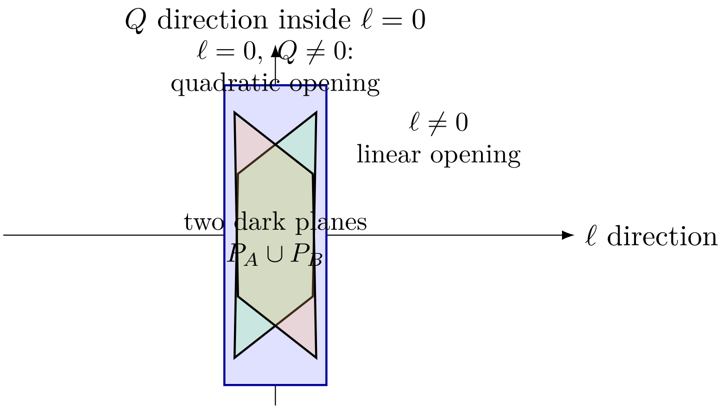

Figure 3: Schematic direction stratification. The ambient hyperplane $\ell = 0$ contains the two exact-dark planes; points elsewhere on the hyperplane have a quadratic opening.

5.1 Certificate discipline

A signed involution S with $SH_0 = H_0S$, $Se_0 = e_0$ and $Se_5 = -e_5$ is sufficient for baseline darkness. Under perturbation the commutator $[S, D(d)]$ may become non-zero. This proves only that this certificate has failed. Exact opening requires a surviving coefficient of N, of the moment series, or of K. Conversely, a point of $V(L, Q)$ may be dark while failing to preserve a chosen presentation of S.

Proposition 5.3 (Coefficient syzygy)**.** *The triple* (L, Q, R) *satisfies the exact relation*

$$d_2d_4L + (d_2 + d_4)Q + R = 0.$$

The relation explains why a nominal cubic numerator coefficient supplies no third stratum after $L = Q = 0$*.*

Proof. Expand the left-hand side and cancel the four cubic monomials. This is equation (11) rewritten and is also KERNEL-CHECKED as `M3A.coefficient_syzygy`. □

This syzygy is the algebraic reason that the fixed-ray classification has only linear, quadratic and exact-zero branches. It is not a statement that the cubic moment itself is generated by the two earlier moments; the tagged depth information is addressed in section 12.

6 Endpoint-access word support

The support restriction is clearest in a rectangular input–output setting. Let

$$A(\delta) = A_0 + \sum_{r=1}^{m} \delta_r V_r, \qquad M_k(\delta) = CA(\delta)^k B.$$

For a multi-index $\gamma \in \mathbb{N}^m$ with $|\gamma| = j$, let $M_{\gamma,k}$ be the coefficient of δ^γ in M_k. Define endpoint depths

$$r_s(r) = \min\{q \geq 0 : V_r A_0^q B \neq 0\}, \qquad r_t(r) = \min\{q \geq 0 : CA_0^q V_r \neq 0\},$$

with value ∞ if the set is empty. If a word realising γ begins with label u at the source end and ends with label v at the target end, call (u, v) compatible. Let

$$\rho(\gamma) = \min_{(u,v) \text{ compatible}} (r_s(u) + r_t(v)).$$

Theorem 6.1 (Endpoint-access bound)**.** *For $j = |\gamma| \geq 1$,*

$$M_{\gamma,k} = 0 \quad \textit{whenever} \quad k < j + \rho(\gamma). \tag{16}$$

At $k = j + \rho(\gamma)$, the coefficient is the sum of all compatible endpoint-minimal words with zero internal gaps. The bound is sharp if and only if that boundary sum is non-zero.

Proof. Expand $A(\delta)^k$ into labelled words. A word contributing to δ^γ has j perturbation letters and $k - j$ copies of A_0. If its first perturbation label at the source end is u, at least $r_s(u)$ copies of A_0 must lie to its right; if its last label at the target end is v, at least $r_t(v)$ copies must lie to its left. Internal gaps are non-negative. Therefore $k - j \geq r_s(u) + r_t(v)$, and minimising gives equation (16). Equality forces endpoint-minimal blocks and zero internal gaps. Summing all such words proves the boundary formula. Distinct words can cancel, so existence of words is not sharpness. □

The fully rectangular multi-index theorem above is PROVED by the labelled-word argument. The present Lean package formalises the locked scalar-ray specialisation needed downstream, not this entire reusable generality.

For the locked model $B = e_0$, $C = e_5^\top$, and $V_r = E_{rr}$ for $r = 1, \ldots, 4$. Every endpoint depth equals 1.

Corollary 6.2 (Six-state support boundary)**.** *For every $\gamma \neq 0$,*

$$[d^\gamma]M_k(d) = 0 \qquad (k < |\gamma| + 2).$$

At the minimal layer $k = j + 2$,

$$\sum_{|\gamma|=j} [d^\gamma]M_{j+2}(d)d^\gamma = d_1^j - d_2^j + d_3^j - d_4^j. \tag{17}$$

Thus all mixed monomials vanish on the boundary.

Proof. The depth statement follows from the theorem. On the boundary, the only word is $H_0D(d)^jH_0$, giving $e_5^\intercal H_0D(d)^jH_0e_0$. Reading the four source and target edge products in equation (2) yields signs $+,-,+,-$, proving equation (17). $\square$

Formal status. For the locked channel, the complete finite word recurrence, the vanishing below $k=j+2$, the boundary word and the signed power sum are KERNEL-CHECKED as `M3A.rayCoeffMatrix`, `M3A.sixState_support_bound`, `M3A.rayMomentCoeff_boundary_matrix` and `M3A.sixState_boundary_ray`. These theorems sum actual word coefficients, so boundary cancellation is already included.

The baseline $j=0$ is deliberately excluded from the theorem: endpoint depths are defined through a perturbation letter, and baseline darkness has its own proof. A finite census of 256 small labelled systems was used only to stress-test common failure modes; it is MEASURED with $n=256$, not a proof of theorem 6.1.

7 Fixed-direction trichotomy

Set $v=(a,b,c,d)\in\mathbb{R}^4$ and $H_v(\varepsilon)=H_0+\varepsilon D(v)$. Define

$$\ell=a-b+c-d,\qquad Q=bd-ac,\qquad R=ab(c-d)+cd(a-b).$$

The ray numerator is

$$N_v(z;\varepsilon)=\varepsilon\ell(z^2+2z)+2\varepsilon^2Q(z+1)+\varepsilon^3R. \tag{18}$$

Theorem 7.1 (Complete fixed-ray trichotomy). *Exactly one of the following holds.*

(a) If $\ell\neq 0$, the first perturbation/time pair is $(p,q)=(1,3)$, and the coefficient of εt^3 in K_v is $i\ell/6\neq 0$. Every other surviving exponent lies in $((1,3)+\mathbb{N}^2)\setminus\{(1,3)\}$.

(b) If $\ell=0$ and $Q\neq 0$, then $(p,q)=(2,4)$, and the coefficient of ε^2t^4 in K_v is $Q/12\neq 0$. Every other surviving exponent lies in $((2,4)+\mathbb{N}^2)\setminus\{(2,4)\}$.

(c) If $\ell=Q=0$, then $K_v(\varepsilon,t)\equiv 0$; there is no cubic opening branch.

Proof. The exact numerator equation (18) first separates the three cases. If $\ell=Q=0$, the syzygy forces $R=0$, so the rational entry and hence the entire amplitude vanish. For the first two cases, the endpoint support theorem identifies the earliest possible time depth. Direct boundary evaluation gives $[\varepsilon]M_3=\ell$ and $[\varepsilon^2]M_4=2Q$. Multiplication by $(-i)^3/3!=i/6$ and $(-i)^4/4!=1/24$ gives the stated amplitude coefficients. Non-zero coefficients prove sharpness. $\square$

Formal status. The literal coefficient support and all three mutually exclusive branches are KERNEL-CHECKED over complex directions by `M3A.linearRayData_of_linear_ne`, `M3A.quadraticRayData_of_linear_zero_quadratic_ne`, `M3A.darkRayData_of_linear_quadratic_zero` and `M3A.fixedRay_trichotomy`. The claim that exactly one branch is selected is additionally traced to `M3A.classifyRay_eq_linear_iff`, `M3A.classifyRay_eq_quadratic_iff`, `M3A.classifyRay_eq_dark_iff` and `M3A.classifyRay_spec`. The exact coefficient values are checked in M5, while M8 identifies the array with the convergent matrix-exponential amplitude. The real statement here follows by scalar restriction.

The ordered pair (p,q) is essential. The numerator valuation p alone does not encode time depth; q alone does not encode perturbative order. For the protected direction $v_*=(1,0,-1,0)$, $\ell=0$, $Q=1$, and the first amplitude coefficient is exactly $(1/12)$.

Corollary 7.2 (Dark rays)**.** *The fixed dark directions are precisely*

$$(a-b, c-d) = (0,0) \quad \text{or} \quad (a-d, c-b) = (0,0).$$

Remark 7.3 (Family-specific completeness)**.** *The absence of a cubic branch is exact for this four-parameter diagonal family. It is not a universal theorem for all analytic Hamiltonian perturbations. A two-state off-diagonal perturbation proportional to ε^3 supplies an immediate cubic-opening control.*

8 Fixed-fibre Newton geometry

For a fixed direction v, let

$$K_v(\varepsilon, t) = \sum_{p,q \geq 0} A_{p,q}(v) \varepsilon^p t^q.$$

Its Newton polyhedron is the convex hull of $\{(p,q) : A_{p,q}(v) \neq 0\} + \mathbb{R}^2_{\geq 0}$. Actual coefficients, rather than candidate support, define this set.

Theorem 8.1 (Fixed-fibre Newton polyhedra)**.** *For the three branches of theorem 7.1, the Newton polyhedron is, respectively,*

$$(1,3) + \mathbb{R}^2_{\geq 0}, \qquad (2,4) + \mathbb{R}^2_{\geq 0}, \qquad \varnothing.$$

Every positive weight exposes the unique displayed vertex in the non-dark branches.

Proof. The trichotomy supplies the vertex and its non-zero coefficient. The endpoint bound shows that every other surviving exponent lies in the corresponding upper orthant. A strictly positive linear functional increases along every non-zero vector in that orthant, hence exposes the vertex uniquely. ☐

Formal status. The definition uses the actual non-zero coefficient support from M6 and the Mathlib real convex hull. The empty-germ convention, the general one-vertex orthant lemma, the three fibre shapes and positive-weight uniqueness are KERNEL-CHECKED as `M3A.newtonPolyhedron_empty`, `M3A.newton_eq_orthant_of_vertex`, `M3A.fixedFibre_newton` and `M3A.fixedFibre_positiveWeight_faces`.

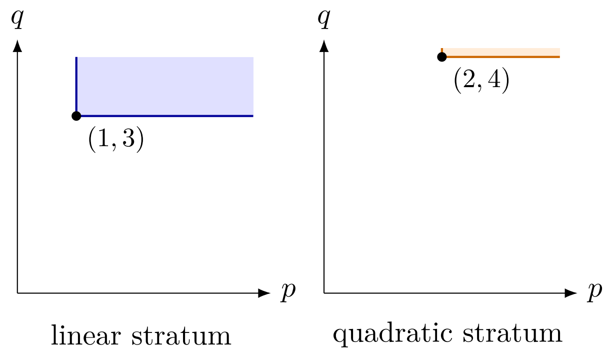

Figure 4: Fixed-fibre Newton polyhedra. The vertex moves from $(1,3)$ to $(2,4)$ on the quadratic stratum; the exact-dark fibre has empty support.

The change of variables $x = z^{-1}$ in corollary 4.3 is not cosmetic. The resolvent expansion is a Laurent series at $z = \infty$; multiplying by x^{-1} aligns the coefficient of x^k with moment depth k. Omitting that shift displaces every Newton point by one and corrupts the comparison with the amplitude.

Classical Newton-polyhedron methods organise dominant exponents and exposed faces (Kouchnirenko, 1976; Varchenko, 1976). Here the fibre theorem is especially simple because there is one lower vertex. The subtlety begins when directions move or coordinates are pulled back, allowing several parent exponents to collide.

Formalisation boundary. This completes the M0–M8 end-to-end Lean spine. Beginning with section 9, the paper moves to conventionally proved extensions supported by exact symbolic or combinatorial checks. No theorem in the remaining sections is represented as part of the current Lean package unless explicitly stated otherwise.

9 Monomial pullback and face collision

This section begins the extension beyond the current M0–M8 formalisation. Its theorems are PROVED by the displayed finite-fibre arguments and checked on released exact examples, but are not yet Lean theorems.

Let $F(x) = \sum_{\alpha\in\mathbb{N}^n} A_\alpha x^\alpha$ be a formal series with coefficients in a finite-dimensional vector space. Fix an exponent matrix $\Lambda \in \mathbb{N}^{m\times n}$ with no zero column and non-zero leading constants c_i. Define

$$x_i = c_i y^{\lambda_i}, \qquad \lambda_i \in \mathbb{N}^m.$$

Theorem 9.1 (Fibre aggregation)**.** *The pulled coefficient at child exponent β is*

$$B_\beta(c) = \sum_{\Lambda\alpha=\beta} A_\alpha c^\alpha. \tag{19}$$

The sum is finite. Hence the actual child support is contained in, but may be strictly smaller than, the naive image $\Lambda(\operatorname{supp} F)$.

Proof. Substitution gives $A_\alpha c^\alpha y^{\Lambda\alpha}$. Group equal child monomials to obtain equation (19). If $\Lambda\alpha = \beta$, then for every column choose a positive entry; this bounds each α_i by a coordinate of β, so the fibre is finite. Its sum can cancel. □

For a positive child weight μ, set $w = \Lambda^\mathsf{T}\mu$ and let ϕ^{gr} denote substitution by leading monomials. This valuation transport is adjacent to tropical elimination (Sturmfels and Tevelev, 2008); the coefficient-fibre cancellation test is kept explicit because equations can retain information lost by set-theoretic support alone (J. Giansiracusa and N. Giansiracusa, 2016).

Theorem 9.2 (Candidate face and equality criterion)**.** *If $\nu_w(F)$ is the parent valuation, then*

$$\nu_\mu(\phi F) \geq \nu_w(F).$$

The candidate component at that weight is

$$\phi^{\mathrm{gr}}(\operatorname{in}_w F). \tag{20}$$

Equality holds if and only if equation (20) is non-zero. A literal identity $\operatorname{in}_\mu(\phi F) = \phi^{\mathrm{gr}}(\operatorname{in}_w F)$ without the non-vanishing hypothesis is FALSIFIED.

Proof. Every parent monomial has child weight $\mu \cdot \Lambda\alpha = w \cdot \alpha$, so no pulled term can have smaller weight. Aggregating all parent terms of minimal weight gives equation (20). If it survives, it is the child initial form. If it cancels, the actual valuation is larger and the displayed right-hand side is zero, whereas the child initial form, when the pullback is non-zero, occurs at a later weight. □

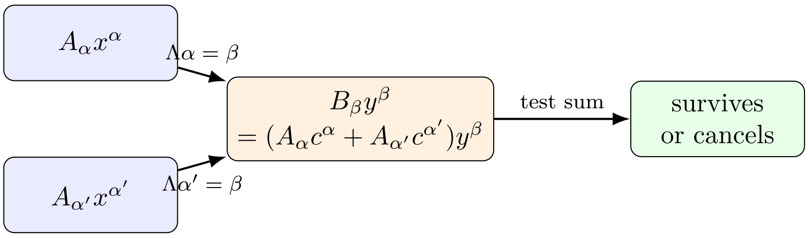

Figure 5: Two parent exponents can enter the same child fibre. Their aggregate, not their separate presence, decides the child support.

Under successive monomial maps with exponent matrices Λ then Ω, exponents compose as $\Omega\Lambda$; leading constants compose as $c_i d^{\lambda_i}$. After a cancellation, however, the next survivor can depend on the full germ and on unit jets, not merely on the first exponent matrix. No universal "next face" rule from the cancelled face alone is claimed.

10 Analytic-arc valuation

Let $v(\eta) = (a(\eta), b(\eta), c(\eta), d(\eta))$ be an analytic germ and set

$$\alpha = \operatorname{ord}_\eta \ell(v(\eta)), \qquad \beta = \operatorname{ord}_\eta Q(v(\eta)),$$

with order ∞ for the zero germ. The three-scale pullback is $\varepsilon = c_\varepsilon s^b$, $t = c_t s^a$, $\eta = c_\eta s^c$, with positive exponents a, b, c and non-zero constants.

Theorem 10.1 (Analytic-arc lower hull)**.** *The only candidate lower points are*

$$P_\ell = (1, 3, \alpha), \qquad P_Q = (2, 4, \beta),$$

omitting a point whose order is ∞. Their pulled costs are

$$\kappa_\ell = b + 3a + c\alpha, \qquad \kappa_Q = 2b + 4a + c\beta.$$

The lower hull is classified as follows:

(i) if $\alpha \le \beta < \infty$, P_ℓ dominates;

(ii) if $\beta < \alpha < \infty$, both points are vertices, with a tie on $c(\alpha - \beta) = a + b$;

(iii) if $\alpha = \infty$ and $\beta < \infty$, only P_Q remains;

(iv) if $\alpha = \beta = \infty$, the arc is exactly dark.

Proof. The endpoint theorem gives $q \ge p + 2$ for every positive perturbation degree p; the $p = 0$ layer is zero by baseline darkness. Every positive-degree coefficient polynomial vanishes on the exact-dark locus. Since (L, Q) is radical by proposition 5.2, each such coefficient belongs to (L, Q). More sharply, complete first-order protection shows that every $p = 1$ coefficient lies in (L). After restriction to the analytic arc, a $p = 1$ term therefore has order at least α and cost at least $b + 3a + c\alpha = \kappa_\ell$.

For $p \ge 2$, coefficient membership in (L, Q) gives order at least $\min(\alpha, \beta)$, while endpoint support gives cost at least

$$pb + (p + 2)a + c\min(\alpha, \beta).$$

If $\alpha \le \beta$, this exceeds κ_ℓ by at least $a+b$. If $\beta < \alpha$, it is at least κ_Q, with equality possible only at the displayed $(2, 4, \beta)$ boundary candidate. Thus no hidden higher-order coefficient can lie below the two stated candidates. Comparing their costs gives $\kappa_\ell - \kappa_Q = c(\alpha-\beta) - (a+b)$, which yields the four cases. If both germs vanish, every positive-degree coefficient vanishes by membership in (L, Q), while the baseline layer is already dark; hence the whole arc is exactly dark. □

On a tie, write $\ell = \ell_\alpha \eta^\alpha + \cdots$ and $Q = Q_\beta \eta^\beta + \cdots$. The candidate amplitude coefficient is

$$\frac{i\ell_\alpha c_\varepsilon c_\eta^\alpha c_t^3}{6} + \frac{Q_\beta c_\varepsilon^2 c_\eta^\beta c_t^4}{12}. \tag{21}$$

Proposition 10.2 (Field-corrected tie rule)**.** *Over $\mathbb{C}$, the tied coefficient can cancel at*

$$c_t = -\frac{2i\ell_\alpha c_\eta^{\alpha-\beta}}{Q_\beta c_\varepsilon}. \tag{22}$$

For a real-analytic Hermitian family with real $\ell_\alpha, Q_\beta, c_\varepsilon, c_\eta, c_t \neq 0$, the two terms in equation (21) have orthogonal real and imaginary parts and cannot cancel. No real non-cancellation statement is made for complex leading data.

Proof. Factor the non-zero cubic term from equation (21); the remaining affine factor in c_t gives equation (22). Under the stated real hypotheses, the first summand is purely imaginary and the second purely real. □

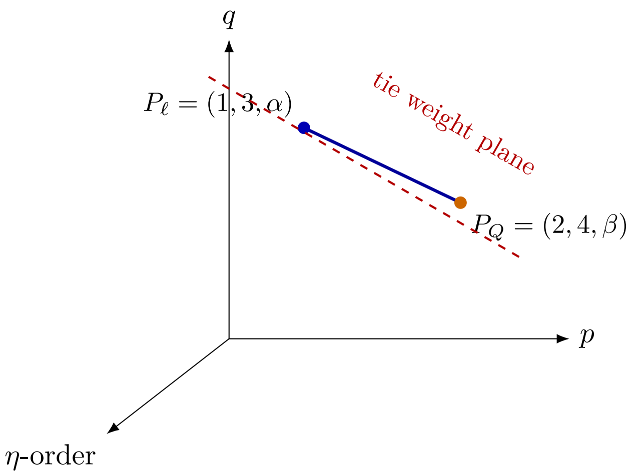

Figure 6: The two analytic-arc candidates. A weight plane can expose one point or their joining edge; coefficient cancellation is a separate test on that edge.

10.1 A tie and its next survivor

Take $v(\eta) = (1+\eta^2, 0, -1, 0)$, so $\alpha = 2$, $\beta = 0$, and use equal scale exponents. At total order six,

$$[s^6]K = \frac{c_\varepsilon^2 c_t^4}{12} + \frac{ic_\varepsilon c_t^3}{6}.$$

With $c_\varepsilon = 1, c_t = -2i$, this cancels. Exact expansion gives

$$[s^7]K = \frac{c_\varepsilon c_t^4}{12} - \frac{ic_\varepsilon^2 c_t^5}{60} = \frac{4}{5}.$$

The survivor $4/5$ is VERIFIED-EXACT. Its value cannot be inferred from (α, β) alone; it uses later coefficients of the pulled germ.

Regular analytic reparametrisation preserves α, β and leading scales. A singular reparametrisation $\eta = \xi^r$ multiplies both finite orders by r. Analyticity is essential: the smooth-flat germ e^{-1/η^2} has zero Taylor series at the origin without vanishing on a punctured neighbourhood.

Finally, if an analytic arc is dark, the product factorisation on $L = 0$ and the integral-domain property of convergent power-series rings show that the whole germ lies in one of the two dark components. It cannot switch components infinitely often near the origin while remaining analytic.

11 Universal multivariate Newton atlas

Return to the actual coordinates $d_1, \ldots, d_4, t$. By corollary 6.2, every non-zero moment monomial $d^\gamma t^k$ with $\gamma \neq 0$ satisfies $k \geq |\gamma| + 2$, and the boundary consists only of pure powers.

Theorem 11.1 (Universal Newton polyhedron)**.** *The Newton polyhedron of $K(d, t)$ at the dark origin is*

$$\operatorname{conv}\{(e_1, 3), (e_2, 3), (e_3, 3), (e_4, 3)\} + \mathbb{R}^5_{\geq 0}. \tag{23}$$

At the four vertices the raw moment coefficients are $+1, -1, +1, -1$, and the amplitude coefficients are $i/6, -i/6, i/6, -i/6$.

Proof. For $\gamma = 0$, all coefficients vanish by the all-depth baseline-darkness theorem, so no unperturbed exponent enters the universal support. The $j = 1$ case of equation (17) supplies the four non-zero vertices. For any other exponent (γ, k) with $\gamma \neq 0$, choose r with $\gamma_r > 0$. Then $k \geq |\gamma| + 2 \geq 3$, and $(e_r, 3) \leq (\gamma, k)$ coordinatewise. Hence all actual support lies in the upper closure of the four vertices. The amplitude coefficients follow from $(-i)^3/3! = i/6$. □

For a positive weight $\omega = (\omega_1, \ldots, \omega_4, \tau)$, the exposed vertices are indexed by the non-empty subset

$$I(\omega) = \{r : \omega_r = \min_j \omega_j\}.$$

Every non-empty subset occurs, giving exactly $2^4 - 1 = 15$ relatively open cells. On a cell I, the face polynomial is

$$\frac{i}{6} t^3 \sum_{r \in I} \sigma_r d_r, \qquad (\sigma_1, \sigma_2, \sigma_3, \sigma_4) = (1, -1, 1, -1). \tag{24}$$

Over the complex torus, every face with at least two vertices has a cancellation hypersurface. Over the positive real torus, a face cancels only when I contains both signs.

11.1 Moment–amplitude face mismatch

The moment and amplitude have the same exponent support, because the multiplier $(-i)^k/k!$ is never zero. Their face cancellation equations need not agree when a pullback makes distinct time depths tie. A reachable two-term face with parent points $(e_1, 4)$ and $(e_2, 3)$ has raw coefficients $2, -1$. Its moment and amplitude face forms are

$$W_M = 2c_1 - c_2, \qquad W_A = \frac{c_1}{12} - \frac{ic_2}{6}.$$

Thus equality of supports does not imply equality of cancellation loci.

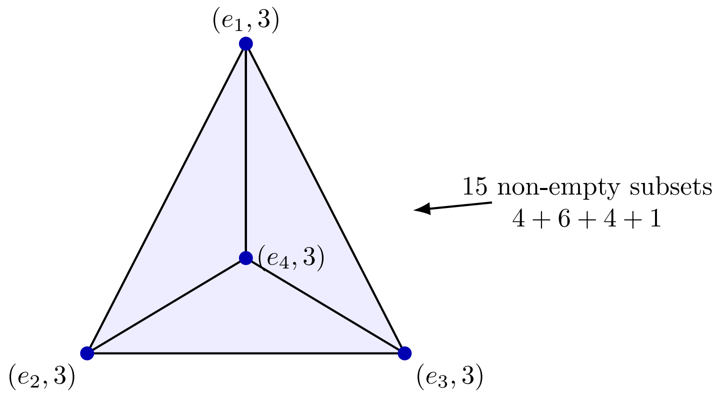

Figure 7: A tetrahedral projection of the four Newton vertices. Its non-empty vertex subsets encode the 15 positive normal cells; this is a combinatorial projection, not a Euclidean embedding of the five-dimensional polyhedron.

11.2 Fixed positive rational weights

The following corrects an overly pessimistic interpretation of the finite universal atlas obstruction.

Theorem 11.2 (Fixed-weight coefficient-stratified termination)**.** *For every fixed positive rational weight, after finite ramification where needed, the pulled amplitude has a finitely generated coefficient ideal, so a finite coefficient-stratified exact-zero/first-survivor decision exists. No effective uniform depth bound or finite global recursive atlas over all positive weights has been proved.*

Proof. Clear denominators in the fixed rational weight by a ramification $s = u^m$. After substituting $d_r = c_r u^{b_r}$ and $t = c_t u^a$, write the pulled amplitude as

$$K_\omega(u; c) = \sum_{n \geq 0} A_n(c) u^n,$$

where each A_n is a polynomial in the leading constants over the base field (or a Laurent polynomial if constants are restricted to the torus). The ring $\mathbb{K}[c_1, \ldots, c_4, c_t]$, and likewise its Laurent localisation, is Noetherian (Atiyah and Macdonald, 1969). Therefore the ideal $J = (A_n : n \geq 0)$ is generated by a finite subset, say with indices at most N. If $A_0(c) = \cdots = A_N(c) = 0$, then every $A_n(c) = 0$, so the pullback is exactly zero. Otherwise its first survivor occurs among those finitely many indices. This proves existence for the fixed weight. The argument neither computes a useful N nor bounds N uniformly as the weight varies. □

The theorem concerns a coefficient-stratified decision, not a weight-only decision. Delayed cancellations can make the necessary depth arbitrarily large across families of weights or germs; that fact is compatible with Noetherianity at each fixed weight.

12 Gröbner, Rees and marked-channel structures

The dark ideal $I = (L, Q)$ records exact zero sets but forgets which generator enters at which perturbation and time depth. The following hierarchy makes the information loss explicit.

Introduce adapted coordinates

$$\lambda = L, \qquad \xi = d_2 - d_3, \qquad \eta = d_4 - d_3, \qquad \zeta = d_3.$$

Then

$$d_1 = \lambda + \xi + \eta + \zeta, \quad d_2 = \xi + \zeta, \quad d_3 = \zeta, \quad d_4 = \eta + \zeta$$

and

$$Q = \xi\eta - \lambda\zeta, \qquad I = (\lambda, \xi\eta) = (\lambda, \xi) \cap (\lambda, \eta). \tag{25}$$

Thus the two components meet normally inside the hyperplane $\lambda = 0$, but their union has codimension two in the ambient four-space; calling it an ambient normal-crossing divisor would be incorrect.

Theorem 12.1 (Gröbner fan of the dark ideal)**.** *In the original coordinates the dark ideal has 16 full-dimensional monomial initial cones. Twelve initial ideals are radical squarefree intersections of two planes. Four are doubled planes:*

$$(d_1, d_3^2), \quad (d_2, d_4^2), \quad (d_3, d_1^2), \quad (d_4, d_2^2).$$

Every cone has dimension 2*, degree* 2*, Hilbert series* $(1+t)/(1-t)^2$ *and Hilbert polynomial* $2n+1$*. Initialisation commutes with the two-component intersection in exactly the twelve radical cones.*

Proof. Run Buchberger reduction for each strict ordering chamber of the four variables, merge chambers with the same reduced marked basis, and retain the 16 distinct initial ideals. Monomial primary decomposition gives the twelve squarefree and four primary cases. Counting standard monomials yields the common Hilbert series and polynomial. The complete marked bases, representative weights, facets and adjacency graph are reproduced in section D. An independent exact script checks the counts and invariants against the released JSON atlas. □

Proposition 12.2 (Rees algebra)**.** *Let* Y_L, Y_Q *mark the two generators. Then*

$$\mathcal{R}(I) \cong \frac{A[Y_L, Y_Q]}{(QY_L - LY_Q)}.$$

The ideal is of linear type and its powers are integrally closed. The blow-up does not separate the two dark components: its exceptional geometry retains a singular section over their intersection.

Proof. The two generators form a regular sequence in the polynomial ring, so the Rees kernel is generated by the Koszul relation $QY_L - LY_Q$. In the adapted normal form $(\lambda, \xi\eta)$, the Newton polyhedron of each monomial power is saturated in its exponent lattice, giving integral closure. The two affine blow-up charts follow by setting either Rees coordinate to one; their equations retain the component-intersection section. □

For an arc with orders $a = \operatorname{ord}\lambda$, $b = \operatorname{ord}\xi$, $c = \operatorname{ord}\eta$, its basic contact profile is

$$\big(\min(a, b+c), \min(a, c), \min(a, b), \min(a, b, c)\big). \tag{26}$$

Arc-space contact loci provide the general geometric context (Ein, Lazarsfeld and Mustaţă, 2004); equation (26) is the exact specialised formula.

12.1 Tagged ideal and marked section

Introduce tags u for perturbation order and x for time depth. The tagged channel ideal is

$$J = (Lux^3,\ Qu^2x^4,\ Ru^3x^5) = (Lux^3, Qu^2x^4), \tag{27}$$

with first syzygy $(Qux, -L)$. Saturating away the tags recovers the dark ideal, but the plain ideal cannot reconstruct the tags.

Let a free module have basis e_L, e_Q, e_R. The numerator marking is

$$s_N = (1 + 2x)e_L + 2(1 + x)e_Q + e_R,$$

and the kernel includes

$$(Qux, -L, 0), \qquad (d_2d_4u^2x^2, (d_2 + d_4)ux, 1).$$

The first records the two-generator relation with depth tags; the second records the cubic syzygy. Even this marked numerator section is not the full amplitude: denominator inversion and factorial phases remain necessary.

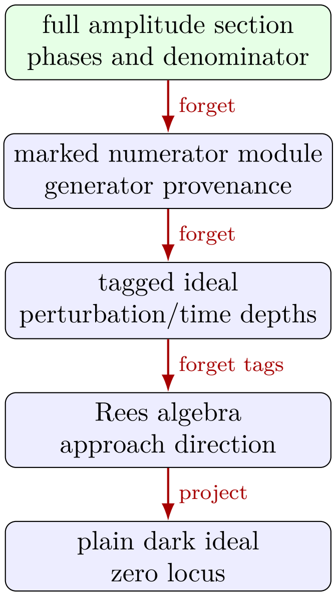

Figure 8: Information hierarchy. Moving left is a lossy forgetful operation; the plain dark ideal cannot recover depth tags or amplitude phases.

For the mixed path $d = (s, s^3, -s, s^4)$, a tied face has

$$W_M = -c^3 + 2c^4, \qquad W_A = -\frac{ic^3}{6} + \frac{c^4}{12}.$$

At $c = 2i$, the amplitude face cancels and the exact next coefficient is $-32/15$ at order seven. This control is VERIFIED-EXACT and exhibits why the full amplitude section is the terminal object in the hierarchy.

13 Counterexamples and scope firewalls

The framework is defined as much by its negative results as by its exact theorems. The following controls block tempting but false upgrades.

Tempting inference	Exact control	Correct conclusion
Broken certificate $\Rightarrow$ opening	A perturbation can destroy a chosen involution while remaining in $V(L,Q)$.	Certificate failure is not a surviving transfer coefficient.
Closure $\Rightarrow$ certificate	Krylov closure is an invariant-subspace property.	It need not supply a signed involution.
Candidate support $\Rightarrow$ child support	Two parent exponents can share a monomial fibre and cancel.	Aggregate by equation (19) first.
Initial form always commutes with pullback	A cancelled candidate face has zero graded pullback but a later non-zero child initial form.	The equality needs the non-vanishing criterion of theorem 9.2.
Moment face = amplitude face	The reachable forms $2c_1 - c_2$ and $c_1/12 - ic_2/6$ have different zeros.	Support agrees; cancellation can differ.
Real tie cannot cancel	Equation equation (22) cancels over $\mathbb{C}$.	Real non-cancellation needs real-analytic Hermitian data and real constants.
The cubic numerator creates a third branch	$R = -(d_2 + d_4)Q - d_2 d_4 L$.	No cubic fixed-ray branch exists in this family.
No cubic opening exists at all	A two-state family with off-diagonal coupling ε^3 opens cubically.	The trichotomy is family-specific.
Plain ideal retains depth	(L,Q) is unchanged after tags are forgotten.	Use the tagged ideal or marked section.
Blow-up separates the two components	The Rees charts retain a singular section over the intersection.	Blow-up records approach data but does not separate these components.
Finite fixed-weight decision $\Rightarrow$ uniform atlas	Noetherianity chooses generators after the weight is fixed.	No effective uniform depth or finite global recursive atlas is proved.
Smooth-flat order ∞ $\Rightarrow$ zero germ	e^{-1/η^2} is flat but non-zero away from 0.	The analytic-arc theorem is not a smooth-germ theorem.
One formalised theorem $\Rightarrow$ the whole paper is formalised	M0–M8 stop at the fixed-fibre analytic bridge.	Use the named theorem map and treat the later extensions as conventionally proved.

Counterexample 13.1 (The symbol R is not the cubic moment)**.** *For $v = (1,0,0,0)$, the cubic numerator coefficient $R(v)$ is zero, while the relevant cubic moment coefficient is* 1*. Algebraic numerator degree and time depth are distinct gradings.*

Counterexample 13.2 (Scalar rays lose the universal fan)**.** *The substitution $d = \varepsilon v$ collapses the four coordinate vertices to a single perturbation coordinate. It therefore cannot retain the 15-cell normal atlas of theorem 11.1. Fixed rays and actual coordinates answer different questions.*

13.1 Publication scope

The exact claims concern the locked matrix equation (2), its four internal diagonal coordinates, formal or analytic local expansions, and declared real or complex coefficient domains. They do not establish robustness under dissipation, experimental noise, arbitrary graph size, arbitrary perturbation support, or a canonical physical interpretation of a multi-Rees construction. Connections to quantum transport, dark states and tropical geometry are positioning links, not priority claims. The priority of the full exact synthesis in the literature is UNAVAILABLE; the prior-art audit found no matching theorem but did not prove absence.

The formal claim is narrower and auditable: only the statements listed in table 1 and section F are KERNEL-CHECKED. This boundary is about proof status, not mathematical importance; it does not weaken the conventional proofs in the extension sections, nor promote their exact scripts to formal proofs.

14 Formal verification, reproducibility and transparency

The release separates kernel checking, conventional proof, exact computation, byte identity and measurement. The formal project is the primary executable evidence for the M0–M8 spine. Exact integer and rational scripts remain independent regression checks for the locked numerator, denominator, moment boundary, arc controls and atlas invariants. No floating-point tolerance enters a theorem certificate.

14.1 Lean build and trust report

The formalisation pins both Lean and Mathlib to version 4.32.1. Its root module imports `M3A.AnalyticBridge`, which transitively imports every earlier stage. A clean `lakebuild` completed successfully after M8. The principal axiom reports contain only `propext`, `Classical.choice` and `Quot.sound`; there is no `sorry`, `admit`, user-declared axiom or unsafe theorem dependency. The theorem-level map is reproduced in section F.

This trust statement follows the standard Lean architecture (de Moura and Ullrich, 2021): tactics and automation may construct proof terms, but acceptance of those terms is checked by the kernel. Mathlib provides the matrix, power-series, topology and convexity infrastructure used in the development (The mathlib Community, 2020). Neither citation is used as evidence for the M3A theorems themselves; their evidence is the released source and checked proof terms.

14.2 Independent exact routes

1. **Lean route.** Finite matrix definitions, cofactor expansion, word recurrence, support classification, convex hull and analytic summation form one checked dependency chain.
2. **Cofactor route.** A direct Leibniz determinant gives (N) and the 52-term denominator.
3. **Moment route.** Polynomial matrix powers give M_k, endpoint support and the signed minimal layer.
4. **Pulled-amplitude route.** Gaussian-rational series compute the ARC1 and GF1 tie cancellations and next survivors.
5. **Combinatorial route.** Released JSON atlases are checked for all 15 MV1 cells and all 16 GF1 cones, including radicality flags and Hilbert data.

The cofactor and moment routes meet at the Laurent expansion of the resolvent; agreement is a cross-check, not circular evidence. Although finite Hamiltonian entries generate highly structured series, no general D-finite closure claim is needed here (Stanley, 1980).

14.3 Executable discrepancy fixture

The fixture changes the symmetric edge pair $H_{0,2} = H_{2,0} = -1$ to $+1$. It then reruns the same cofactor and moment code. Both routes detect disagreement with the locked numerator and moment layer. The fixture result is computed at run time and written to `DATA/discrepancy_detection.json`; it is not a stored truth value.

14.4 Source recovery

The workstation scan hashed 3,636 scientific-source files and found 932 unique byte streams, including 631 duplicate-content groups. It searched 780 unique text streams and extracted 20 unique PDFs without a read failure. Those counts are MEASURED with the stated sample sizes. Vendor TeX distributions, dependency caches, rendered scratch files and the output tree were excluded and recorded. Byte-identical copies were searched once and linked by SHA-256; filename suffixes such as "copy" or parenthetical numbers were not treated as authority.

Paper I has the verified Zenodo version DOI 10.5281/zenodo.20556571 and is cited accordingly (Medford, 2026b). Its source is available in the public GitHub repository. The Zenodo record is version `v1.0.0`; as of 10 August 2026 the GitHub repository has no separate Release or tag. Paper II reproducibility materials are located under `subsequent-work/paper-2-opening-theory/` in that repository. The earlier R1.2 Paper II manuscript remains an author-supplied unpublished preprint with no invented identifier.

14.5 How to reproduce

From the formalisation root, run

```
lake build
lake env lean M3AFormalisation.lean
```

From `subsequent-work/paper-2-opening-theory`, run

```
python CODE/verify_paper.py
python CODE/verify_paper.py --fixture
cd paper/source
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The paper build uses Biber for Harvard author–date citations. Exact environment versions, milestone verification reports, expected JSON and the release manifest accompany the source. The full denominator, atlases and claim maps are machine-readable rather than transcribed by hand.

14.6 Known reproducibility boundary

The current Lean package is complete through the fixed-fibre analytic theorem, but not through the paper's later pullback, arc, atlas, Gröbner/Rees or Noetherian sections. Their exact verifiers check formulas, examples and finite atlases; they do not turn those conventional proofs into kernel-checked ones. Separately, the fixed-weight theorem proves finite generation rather than an effective uniform bound. Its verifier checks hypotheses and examples but cannot turn a non-effective Noetherian argument into a global finite atlas. Both limitations are reported rather than hidden.

15 Conclusion

The fixed-ray opening theory of the locked six-state channel is now kernel-checked from the matrix entries to the analytic amplitude. Lean verifies the signed involutions and all-depth baseline darkness, the exact unreduced numerator, the pointwise union of the two dark planes, the locked endpoint word support, the complete linear/quadratic/exact-zero trichotomy, the two fixed-fibre Newton orthants and the globally convergent matrix-exponential bridge. In particular, the no-cubic branch conclusion concerns the actual selected amplitude, not merely a formal numerator.

The mathematics is organised by a small algebraic core with non-interchangeable representations. The numerator has three displayed coefficients, but its syzygy reduces exact darkness to $L = Q = 0$. Endpoint words bound where coefficients may appear, yet boundary words can cancel. Moment and amplitude supports coincide, while factorial phases can change a pulled-face cancellation equation. Actual coefficient support, rather than candidate words, is what enters the Newton polyhedron.

The later sections extend that verified core without being folded into its formal status. Monomial maps require aggregation in child fibres; analytic arcs require a field-corrected tie rule; universal coordinates produce the four-vertex polyhedron and 15-cell positive normal atlas; and the plain dark ideal, tagged ideal, Rees data and marked section retain progressively more approach and depth information. For each fixed positive rational weight, Noetherianity gives a finite coefficient-stratified decision, but neither an effective uniform depth bound nor a finite global recursive atlas.

The next formal task is therefore well defined: extend the Lean development from the locked scalar-ray endpoint theorem to the general labelled multi-index statement, then formalise monomial-fibre aggregation and the moving-arc valuation theorem. The next mathematical task is to identify hypotheses under which the fixed-weight generating set can be bounded from the exponent matrix, channel dimension and numerator degree. Both directions now start from an explicit, reproducible theorem boundary.

A Full denominator audit

The unreduced denominator is $\Delta(z; d) = \det(zI - H(d))$. The exact determinant contains 52 monomials. The ordering below is by total degree and exponent tuple (z, d_1, d_2, d_3, d_4); together the rows are the complete polynomial, not a truncation.

Index	Coefficient	Monomial
1	-8	d_4
2	-8	d_3
3	-8	d_2
4	-8	d_1
5	32	z
6	4	$d_3 d_4$
7	4	$d_2 d_3$
8	4	$d_1 d_4$
9	4	$d_1 d_2$
10	-12	$z d_4$
11	-12	$z d_3$
12	-12	$z d_2$
13	-12	$z d_1$
14	32	z^2

Index	Coefficient	Monomial
15	-4	zd_2d_4
16	-4	zd_1d_3
17	4	z^2d_4
18	4	z^2d_3
19	4	z^2d_2
20	4	z^2d_1
21	-8	z^3
22	2	$zd_2d_3d_4$
23	2	$zd_1d_3d_4$
24	2	$zd_1d_2d_4$
25	2	$zd_1d_2d_3$
26	-5	$z^2d_3d_4$
27	-5	$z^2d_2d_4$
28	-4	$z^2d_2d_3$
29	-4	$z^2d_1d_4$
30	-5	$z^2d_1d_3$
31	-5	$z^2d_1d_2$
32	8	z^3d_4
33	8	z^3d_3
34	8	z^3d_2
35	8	z^3d_1
36	-12	z^4
37	1	$z^2d_1d_2d_3d_4$
38	-1	$z^3d_2d_3d_4$
39	-1	$z^3d_1d_3d_4$
40	-1	$z^3d_1d_2d_4$
41	-1	$z^3d_1d_2d_3$
42	1	$z^4d_3d_4$
43	1	$z^4d_2d_4$
44	1	$z^4d_2d_3$
45	1	$z^4d_1d_4$
46	1	$z^4d_1d_3$
47	1	$z^4d_1d_2$
48	-1	z^5d_4
49	-1	z^5d_3
50	-1	z^5d_2
51	-1	z^5d_1
52	1	z^6

At $d = 0$, the denominator is

$$z^6 - 12z^4 - 8z^3 + 32z^2 + 32z.$$

After $z = x^{-1}$ and multiplication by x^6, the constant term is 1. This proves the unit assertion used in corollary 4.3. The numerator table has 16 monomials and is exactly the expansion of equation (9); it is stored in `DATA/MV1_PRIMARY_EXACT.json` together with this denominator.

The cofactor and determinant were recomputed from the locked matrix by a dependency-free Leibniz expansion. The result is VERIFIED-EXACT. No root cancellation assumption is used in the dark-locus proof.

B Endpoint words and pullback proof details

B.1 Labelled word formula

For a word containing j perturbation letters with ordered labels $i_1, \ldots, i_j$, write its A_0-gaps as $q_0, \ldots, q_j$. Its contribution is

$$CA_0^{q_j} V_{i_j} A_0^{q_{j-1}} \cdots V_{i_2} A_0^{q_1} V_{i_1} A_0^{q_0} B, \qquad \sum_{r=0}^{j} q_r = k - j.$$

Summing over label sequences with content γ and over all gap compositions gives $M_{\gamma,k}$. Endpoint depth imposes $q_0 \geq r_s(i_1)$ and $q_j \geq r_t(i_j)$. Equality in the support bound sets every internal q_r to zero and selects an endpoint-minimal pair.

For the M3A channel, $V_r = e_r e_r^\top$, $V_r H_0 e_0 = e_r \neq 0$, and $e_5^\top H_0 V_r = \sigma_r e_r^\top \neq 0$. Thus all endpoint depths are one. At equality the word reduces to $e_5^\top H_0 D^j H_0 e_0$, giving equation (17).

B.2 Finite pullback fibres

The no-zero-column hypothesis in theorem 9.1 cannot be omitted. If column i of Λ is zero, then $\Lambda(\alpha + ne_i) = \Lambda\alpha$ for all n, so a formal child coefficient may receive infinitely many parent terms. With every column non-zero, choose $r(i)$ with $\Lambda_{r(i),i} > 0$. If $\Lambda\alpha = \beta$, then

$$0 \leq \alpha_i \leq \frac{\beta_{r(i)}}{\Lambda_{r(i),i}},$$

which bounds the fibre coordinatewise.

B.3 Why the literal initial-form identity fails

Take $F = x_1 - x_2 + x_1^2$ and pull back $x_1 = x_2 = y$. For the parent weight $w = (1,1)$, $\mathrm{in}_w F = x_1 - x_2$, whose graded pullback is zero. The actual pullback is y^2, so its child initial form is y^2, not zero. This is the minimal exact counterexample to an unconditional identity. It also shows why the next survivor cannot be read from the cancelled face alone.

C Analytic-arc series and mismatch controls

For the ARC1 control $d = (s + s^3, 0, -s, 0)$ and $t = c_t s$, exact Gaussian-rational matrix expansion gives

$$[s^6]K = \frac{c_t^4}{12} + \frac{ic_t^3}{6},$$
$$[s^7]K = \frac{c_t^4}{12} - \frac{ic_t^5}{60}.$$

At $c_t = -2i$, these are 0 and $4/5$. The calculation uses the full matrix exponential series through depth seven; it is not obtained by assuming the claimed survivor.

For the GF1 mixed path $d = (s, s^3, -s, s^4)$, $t = c_t s$, the first two tied raw moment contributions produce

$$[s^6]K = -\frac{ic_t^3}{6} + \frac{c_t^4}{12},$$
$$[s^7]K\big|_{c_t=2i} = -\frac{32}{15}.$$

The first expression vanishes at (2i). These two controls use opposite signs of the imaginary time scale and are not copies of one another.

The reachable moment–amplitude mismatch can be summarised as follows.

Parent points	Moment face	Amplitude face
$(e_1, 4), (e_2, 3)$	$2c_1 - c_2$	$c_1/12 - ic_2/6$
ARC1 tie	$2c_\varepsilon^2 c_t^4 + \cdots$	$c_\varepsilon^2 c_t^4/12 + ic_\varepsilon c_t^3/6$
GF1 mixed path	$-c^3 + 2c^4$	$-ic^3/6 + c^4/12$

The first row proves the logical point most cleanly: non-zero scalar conversion at each exponent preserves support, while unequal k-dependent conversions change the face equation.

D The 15-cell and 16-cone atlases

D.1 MV1 positive normal cells

The table lists every non-empty subset of the four universal Newton vertices. The face form omits the common factor $it^3/6$. "Positive cancel" means a zero exists with every leading coordinate strictly positive and real.

Cell	I	Face form (without $it^3/6$)	$\mathbb{C}^*$ cancel	$\mathbb{R}_{>0}$ cancel
1	$\{1\}$	c_1	no	no
2	$\{2\}$	$-c_2$	no	no
3	$\{3\}$	c_3	no	no
4	$\{4\}$	$-c_4$	no	no
5	$\{1,2\}$	$c_1 - c_2$	yes	yes
6	$\{1,3\}$	$c_1 + c_3$	yes	no
7	$\{1,4\}$	$c_1 - c_4$	yes	yes
8	$\{2,3\}$	$-c_2 + c_3$	yes	yes
9	$\{2,4\}$	$-c_2 - c_4$	yes	no
10	$\{3,4\}$	$c_3 - c_4$	yes	yes
11	$\{1,2,3\}$	$c_1 - c_2 + c_3$	yes	yes
12	$\{1,2,4\}$	$c_1 - c_2 - c_4$	yes	yes
13	$\{1,3,4\}$	$c_1 + c_3 - c_4$	yes	yes
14	$\{2,3,4\}$	$-c_2 + c_3 - c_4$	yes	yes
15	$\{1,2,3,4\}$	$c_1 - c_2 + c_3 - c_4$	yes	yes

D.2 GF1 Gröbner cones

The variables in this table are abbreviated $a = d_1, b = d_2, c = d_3, d = d_4$. Representative weights are strict, and adjacency is across a codimension-one wall. The full JSON also stores reduced bases, facets, primary decompositions and component-initial comparisons.

Cone	Weight (a, b, c, d)	Initial ideal	Radical	Adjacent
C01	$(4, 3, 2, 1)$	(a, bc)	yes	C02, C03, C05
C02	$(4, 3, 1, 2)$	(a, bd)	yes	C01, C04, C06, C13
C03	$(4, 2, 3, 1)$	(a, c^2)	no	C01, C04, C09
C04	$(4, 1, 2, 3)$	(a, cd)	yes	C02, C03, C14
C05	$(3, 4, 2, 1)$	(b, ac)	yes	C01, C06, C07, C10

Cone	Weight (a,b,c,d)	Initial ideal	Radical	Adjacent
C06	$(3,4,1,2)$	(b,ad)	yes	C02, C05, C08
C07	$(1,4,3,2)$	(b,cd)	yes	C05, C08, C12
C08	$(2,4,1,3)$	(b,d^2)	no	C06, C07, C15
C09	$(3,2,4,1)$	(c,a^2)	no	C03, C10, C11
C10	$(2,3,4,1)$	(c,ab)	yes	C05, C09, C12
C11	$(2,1,4,3)$	(c,ad)	yes	C09, C12, C14
C12	$(1,3,4,2)$	(c,bd)	yes	C07, C10, C11, C16
C13	$(3,2,1,4)$	(d,ab)	yes	C02, C14, C15
C14	$(3,1,2,4)$	(d,ac)	yes	C04, C11, C13, C16
C15	$(2,3,1,4)$	(d,b^2)	no	C08, C13, C16
C16	$(1,2,3,4)$	(d,bc)	yes	C12, C14, C15

The four non-radical rows are $C03, C08, C09, C15$. They are precisely the orderings in which the leading linear variable and the square in the remaining quadratic generator select the same plane. Their radicals agree with one dark component, but their doubled structures preserve the common degree two.

E Rees charts, contact profiles and the marked module

Let $A = \mathbb{K}[\lambda, \xi, \eta, \zeta]$ and $I = (\lambda, \xi\eta - \lambda\zeta)$. The Rees presentation is

$$A[Y_L, Y_Q]/((\xi\eta - \lambda\zeta)Y_L - \lambda Y_Q).$$

On the $Y_L \neq 0$ chart, set $r = Y_Q/Y_L$; the equation is $\xi\eta = \lambda(\zeta + r)$. On $Y_Q \neq 0$, set $s = Y_L/Y_Q$; the equation is $s(\xi\eta - \lambda\zeta) = \lambda$. Above $\lambda = \xi = \eta = 0$, both charts retain a positive-dimensional section, which is why the blow-up does not separate the two components.

For an analytic arc, write $a = \operatorname{ord}\lambda, b = \operatorname{ord}\xi, c = \operatorname{ord}\eta$. The orders of the union ideal and the two component ideals are

$$\operatorname{ord} I = \min(a, b+c), \quad \operatorname{ord}(\lambda,\eta) = \min(a,c), \quad \operatorname{ord}(\lambda,\xi) = \min(a,b).$$

Adding the intersection order $\min(a,b,c)$ gives equation (26).

The tagged ideal lives in $A[u,x]$. Its two-generator presentation

$$A[u,x]^2 \longrightarrow J, \qquad (f,g) \longmapsto fLux^3 + gQu^2x^4$$

has kernel generated by $(Qux, -L)$. Adding a formal R-basis vector produces a three-generator marked module with the second kernel vector displayed in section 12. The section s_N keeps the resolvent-polynomial marking. To obtain the amplitude section one must also multiply by the inverse denominator series and by the depth-dependent phases $(-i)^k/k!$.

There is no claim that this marked module is a canonical physical state space. It is an exact algebraic bookkeeping device for coefficient provenance.

F Lean traceability and release checks

F.1 Kernel-checked claim map

Paper claim	Lean theorem	Dependency role
Baseline moments vanish at every depth	`baseline_moments_dark`	Signed-certificate base case
Exact selected numerator equation (9)	`selectedNumerator_formula`	Universal cofactor identity
Pointwise dark-plane union equation (13)	`ell_quad_zero_iff_planes`	Selects a persistent certificate
Locked support below $k = j + 2$	`sixState_support_bound`	Excludes premature time layers
Signed boundary power sum equation (17)	`sixState_boundary_ray`	Gives the first candidate coefficients
Complete first-order vanishing on $L = 0$	`amplitudeCoeff_one_eq_zero_of_linear_zero`	Excludes all $p = 1$ terms
Fixed-ray classifier equivalences	`classifyRay_eq_linear_iff`, `classifyRay_eq_quadratic_iff`, `classifyRay_eq_dark_iff`, `classifyRay_spec`	Exactly one branch predicate matches the classifier
Fixed-ray trichotomy theorem 7.1	`fixedRay_trichotomy`	Exhaustive support classification
Fixed-fibre Newton theorem theorem 8.1	`fixedFibre_newton`	Convexifies actual coefficient support
Positive-weight exposed vertices	`fixedFibre_positiveWeight_faces`	Unique minimiser in each non-dark fibre
Analytic coefficient bridge equation (1)	`selectedRayAmplitude_eq_tsum_coefficients`	Identifies support with the matrix exponential
Exact-dark analytic rays	`selectedRayAmplitude_eq_zero_of_linear_quadratic_zero`	Closes the no-cubic conclusion

The final M8 build completed successfully with 8,669 build jobs reported by Lake. For every principal theorem, `#printaxioms` returned only `propext`, `Classical.choice` and `Quot.sound`. The source contains no admitted proof or user-declared axiom. The scheme-theoretic ideal equality, fully general endpoint theorem and all post-fixed-fibre results are deliberately absent from this table.

F.2 Independent exact-computation map

Claim	Exact route	Released evidence
Selected numerator and denominator	Leibniz determinant	`EXPECTED/paper_verification.json`
Endpoint support and signed boundary	Polynomial matrix powers	same JSON
ARC1 4/5 survivor	Gaussian-rational pulled amplitude	same JSON
GF1 −32/15 survivor	Gaussian-rational pulled amplitude	same JSON

Claim	Exact route	Released evidence
15 positive normal cells	non-empty subset enumeration	`DATA/MV1_15_CELL_ATLAS.json`
16 Gröbner cones	exact atlas invariant check	`DATA/GF1_16_CONE_ATLAS.json`
Perturbed edge detected	recomputed cofactor and moments	`DATA/discrepancy_detection.json`
Source recovery counts	recursive SHA-256/text/PDF scan	`DATA/FULL_SOURCE_SCAN_SUMMARY.json`

The release manifest records SHA-256 for every payload file except the manifest itself. Fresh extraction is checked by copying the compact archive to a new directory, rerunning the exact modes and rebuilding the paper. Visual quality is checked from rasterised pages after a successful PDF build; textual compilation alone is not treated as layout verification.

The random small-system endpoint census has $n = 256$. It is retained only as a diagnostic methodology and labelled MEASURED; the general theorem rests on the labelled-word proof and the locked specialisation rests on its Lean proof.

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